

REVIEW

PCPE1 and PCPE2: When Sequence Similarity Masks Functional Diversity

Mary G. Sorci-Thomas¹, Alexander Rocksvold¹, Bilal Ahmad¹

ABSTRACT: PCPE1 (procollagen C-endopeptidase enhancer 1) and PCPE2 are ECM (extracellular matrix) glycoproteins that lack intrinsic protease activity and regulate procollagen processing. Initially, these glycoproteins were found to be structurally related and assumed to play similar roles in enhancing collagen maturation. Recent evidence suggests that PCPE1 and PCPE2 exert both overlapping and distinct biological functions beyond collagen processing. PCPE1, first identified and shown to enhance collagen assembly in the ECM, now appears to be involved in adipose-to-liver crosstalk, angiogenesis, and fibrosis/wound healing. By comparison, PCPE2, with $\approx 45.6\%$ amino acid identity ($\approx 80.6\%$ amino acid similarity) to PCPE1, inhibits PCPE1's enhancement of procollagen processing while also participating in cellular plasticity, innate immune responses, and adipose tissue expansion. Together, the evidence suggests that PCPE1 and PCPE2 exhibit functional overlap but also play additional regulatory roles beyond traditional ECM remodeling.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: atherosclerosis ■ coronary vessels ■ membrane proteins ■ pathologists ■ tissue distribution

It is a great honor to receive this prestigious award from the American Heart Association, named for the distinguished pathologist and medical educator Dr George Lyman Duff, a pioneer in atherosclerosis research. Since 1956, the American Heart Association Council of *Arteriosclerosis, Thrombosis, and Vascular Biology* has bestowed this award annually in his honor. I (Mary G. Sorci-Thomas) am sincerely grateful for this year's recognition and the opportunities it brings. My research has traditionally focused on the structure and function of HDL (high-density lipoprotein) apoA-I (apolipoprotein A-I) and its role in protecting against atherosclerosis progression.^{1–3} However, several years ago, an intriguing side project piqued my interest.^{4,5} Since then, my laboratory has been keenly interested in the link between cardiometabolic disease and the adipose tissue ECM (extracellular matrix)–resident glycoproteins PCPE1 (procollagen C-endopeptidase enhancer 1) and PCPE2.

Regulation of collagen processing and deposition in the ECM is essential for cellular repair and tissue homeostasis.^{6–8} Defects in procollagen processing can lead to lethal disorders. For example, excessive collagen accumulation in the ECM leads to fibrosis and aberrant

wound repair, thereby altering normal organ function in the liver, kidney, skin, heart, and lung.^{9–11} For collagen to be properly processed, the C-terminal propeptides of procollagens must be enzymatically removed before fibrillar self-assembly can occur.¹² The principal proteinases that cleave these propeptides are collectively known as metalloproteinases and include BTPs (bone morphogenetic protein 1/tolloid-like proteinases).^{11,13} The activity of BTPs on fibrillar collagens is significantly regulated by factors required for optimal cleavage; 2 such cofactors or enhancers are PCPE1 and PCPE2.¹⁴ PCPE1 is known to bind only to procollagen through the C-propeptide region, while PCPE2 binds both BMP1 (bone morphogenetic protein 1) and the C-terminal propeptide of procollagens. Because both PCPE1 and PCPE2 share notable structural homology,¹⁵ it has been assumed that they perform similar biological functions in collagen processing.^{14,16} However, emerging evidence indicates that, despite their structural homology, PCPE1 and PCPE2 exert distinct and, in some cases, opposing effects on procollagen processing and tissue remodeling.¹⁶ PCPE1 is abundantly expressed in collagen-rich tissues, including tendons, skin, bone, and cornea. It

Correspondence to: Mary G. Sorci-Thomas, PhD, FAHA, Department of Medicine, Division of Endocrinology and Molecular Medicine, Medical College of Wisconsin, 8701 Watertown Plank Rd, Milwaukee, WI 53226. Email mstomas@mcw.edu

This manuscript was sent to M. Luisa Iruela-Arispe, Guest Editor, for review by expert referees, editorial decision, and final disposition.

© 2026 American Heart Association, Inc.

Arterioscler Thromb Vasc Biol is available at www.ahajournals.org/journal/atvb

Nonstandard Abbreviations and Acronyms	
BMP1	bone morphogenetic protein 1
BTP	bone morphogenetic protein 1/tolloid-like proteinase
ECM	extracellular matrix
HDL	high-density lipoprotein
HSPG	heparan sulfate proteoglycan
KO	knockout
LDLR	low-density lipoprotein receptor
PCPE	procollagen C-endopeptidase enhancer
SR-BI	scavenger receptor class B type 1

is present at lower levels in tissues with relatively low collagen I content, including the kidney, heart, and skeletal muscle,^{17–19} and its action is entirely dependent on binding to the C-propeptide region of fibrillar procollagen.²⁰ The enhancing activity of PCPE1 is regulated by domains called Complement C1r/C1s, UEGF, BMP1, or CUB domains,¹⁷ which facilitate optimal substrate presentation to BMP1 for cleavage.²¹ CUB domains are ≈110-residue structural motifs involved in protein-protein interactions²² and occur exclusively in ECM-associated and plasma membrane-associated proteins. Like PCPE1, PCPE2 also contains 2 CUB domains, coordinated by calcium²³; these domains are well known for mediating protein-protein interactions.²⁴ Following the N-terminal CUB domains is a linker domain whose length and amino acid composition vary substantially between PCPE1 and PCPE2.¹⁵ Following the linker domain region is a netrin-like domain, which has been shown to anchor the glycoprotein to HSPGs (heparan sulfate proteoglycans) in the ECM.^{25–27} Although PCPE1 and PCPE2 share structural similarity, there are many ways in which they differ, as listed in the Table. First, the abundance and tissue distribution of PCPE1 differ from those of PCPE2, with PCPE2 mainly expressed

Highlights
• PCPE1 (procollagen C-endopeptidase enhancer 1) and PCPE2 show 45.6% amino acid homology, yet display distinct tissue expression. • Both PCPEs are highly expressed in adipose tissue residing in the extracellular matrix. • Unlike PCPE1, PCPE2 regulates lipid metabolism, likely through regulation of receptor function.

in the artery, adipose tissue, and heart.²⁹ This is largely due to PCPE2 levels being generally lower than those of PCPE1. Interestingly, in an absolute sense, PCPE1 and PCPE2 mRNA abundances in the coronary artery are similar,²⁸ while PCPE1 is about 2-fold higher in adipose tissue compared with PCPE2. Second, PCPE1 has fewer posttranslational modifications and exhibits lower affinity for heparin,^{25,29,34} as shown by a number of experiments that indicate that glycosylation of the PCPE2 consists solely of sialylated Galβ(1–3)GalNAc O-linked cores.²⁹ Third, PCPE1 enhances procollagen processing by promoting BMP1-mediated cleavage of fibrillar procollagens, whereas PCPE2 has been shown to uniquely inhibit BMP1-mediated cleavage at sufficiently high concentrations.^{14,16} These differences suggest that in a cellular context, PCPE1 and PCPE2 do perform divergent functions.

PCPE1 AND PCPE2: HISTORICAL PERSPECTIVES AND RELATED FUNCTION(S)

PCPE1 was first identified in 1986 as a contaminant during the purification of procollagen C-proteinase, where it was observed to enhance the C-terminal processing of procollagen I.^{1,7,35} This discovery provided evidence that the maturation of procollagens is not solely governed by proteolytic enzymes but also regulated by

Table. Summary of Features for PCPE1 and PCPE2

Characteristics	PCPE1	PCPE2
Tissue distribution	Highly expressed in connective tissue, fibroblasts, bone, tendon, skin with lower levels in heart, and adipose tissue ^{9–11,17,18,28}	Generally expressed at lower levels than PCPE1, with heart and adipose tissue showing the highest levels ^{15,28,29}
Glycosylation	N-linked oligosaccharide often decorated with sialic acid and limited complex O-linked oligosaccharides ^{29,30}	O-linked glycan often sialylated little to no N-glycosylation ^{14,15,29}
Metabolic roles	Collagen processing wound repair, adipose-to-liver crosstalk, and angiogenesis ^{25,31}	Receptor function, adipose physiology, ³² innate immune responses, and lipid metabolism ³³
Interaction with receptor SR-BI	No information	Enhances SR-BI-mediated HDL uptake ³²
Cofactor or enhancer activity	Enhances BMP1-mediated cleavage of procollagen C-propeptides ¹⁶	Inhibits BMP1 activity of procollagen C-propeptides ¹⁶
Disease connection	Liver fibrosis and cancer ¹⁵	Cardiovascular disease and cancer ¹⁵

BMP1 indicates bone morphogenetic protein 1; HDL, high-density lipoprotein; PCPE1, procollagen C-endopeptidase enhancer 1; PCPE2, procollagen C-endopeptidase enhancer 2; and SR-BI, scavenger receptor class B type I.

an enhancer that fine-tunes processing. This discovery, therefore, provided a conceptual advance in understanding how fibrillar collagen is processed in the ECM. The gene for PCPE1 (*PCOLCE*) is located on chromosome 7, proximal to the collagen $\alpha 2(I)$; *COL1A2* gene and encodes a ≈ 55 -kDa glycoprotein for humans and rodents, respectively.¹⁸ Structurally, PCPE1 is an elongated multidomain protein ≈ 150 Å in length, which increases the efficiency of BMP1-mediated cleavage of procollagen by up to 20-fold.³⁶ Importantly, PCPE1 exhibits a high degree of substrate specificity by enhancing BMP1-mediated cleavage of fibrillar procollagens while not affecting the processing of other BMP1 substrates.³⁶ As noted earlier, PCPE1 comprises 2 N-terminal CUB domains, a variable linker region, and a C-terminal netrin-like domain.³⁷ The complete CUB domains are both necessary and sufficient for enhancing activity, underscoring their central role in PCPE1 function.^{20,38} Among the most important steps in fibrillar collagen formation is the C-terminal proteolytic processing of the procollagen precursors.¹¹ Although PCPE1 does not act directly on BMP1, it dramatically enhances its activity by binding to the procollagen substrate and inducing conformational changes that render the cleavage site more accessible.¹¹ This substrate-mediated enhancement positions PCPE1 as a potential target for regulating BTP activity and, ultimately, collagen deposition in the ECM.²¹ Mechanistically, the 2 CUB domains are critical for PCPE1 function, as they bind to the C-propeptide trimer of procollagens and promote the C-terminal maturation of collagen type I, II, and III procollagens.^{17,21} Previous structural and biochemical studies have shown that the CUB domains act cooperatively, allowing PCPE1 to bind simultaneously to multiple chains of the procollagen C-propeptide trimer,²¹ also supporting the idea that CUB domains are essential.^{39,40} In addition to PCPE1's role in procollagen processing, it has been reported to form a transient complex with β -amyloid and exhibit high-affinity interactions with laminin-111, collagen VI, endostatin, and collagen IV.³⁴ These interactions suggest that PCPE1 participates in biological processes beyond classical ECM assembly, including neurodegenerative disorders, tumor progression, and angiogenesis.³⁴ Taken together, an understanding of non-ECM-related functions of PCPE1 may reveal previously unrecognized roles across diverse disease contexts, positioning PCPE1 as a promising therapeutic target.

PCPE2, in contrast to PCPE1, is less well studied²⁹ and was identified as a collagen-binding protein mapped to chromosome 3 in humans and chromosome 9 in mice. PCPE2 shares $\approx 45.6\%$ structural similarity with PCPE1,¹⁵ with high conservation within the 2 CUB domains, while the linker and netrin-like domain exhibit much lower sequence homology.¹⁵ Interestingly, the amino acid sequence of PCPE2 in humans compared with mice

is high, sharing $\approx 89.1\%$ amino acid homology.¹⁵ Overall, PCPE2's structure remains highly conserved down the phylogenetic tree, with 81.5% amino acid homology between its entire amino acid sequences when comparing human with coelacanth.¹⁵ PCPE1 shows greater amino acid divergence among species, with only 51.1% amino acid conservation between human and coelacanth.¹⁵ In addition to these differences, PCPE2 exhibits a glycosylation pattern markedly distinct from that of PCPE1, suggesting divergence in posttranslational modification between the 2 proteins.²⁹ Previously reported posttranslational modifications of PCPE2 include both glycosylation and phosphorylation, which occur in a context-dependent manner.¹⁵ One study identified 8 potential phosphorylation sites in PCPE2,⁴¹ including serine phosphorylation within the CUB domain in human Jurkat T-cell leukemia cells⁴² and phosphorylation within the netrin-like domain when isolated from Chang liver cells.⁴³ Moreover, phosphorylation of serine and tyrosine residues in human PCPE2 has also been reported in ovarian and breast cancer xenograft tissues.^{15,44} PCPE2 specifically inhibits BTPs and also interferes with the PCP activity of meprins by interacting with procollagens.¹⁶ In this study, the authors found that PCPE1 has no effect on the BTP-dependent cleavages, while PCPE2 acts as a direct and potent inhibitor of these cleavages. Their results indicate that the CUB domains of PCPE2 interact with BMP1 to allosterically inhibit the activity of the protease domain. In the case of fibrillar procollagens, PCPE1 enhances the C-terminal maturation of procollagens I to III through the binding of CUB1CUB2 to C-propeptides. PCPE2 binds procollagens and BMP1 with similar affinities but cannot interact with both partners.¹⁶ These emerging functional differences highlight the unique regulatory roles of PCPE2 in the extracellular and pericellular environments. High conservation in domain amino acid composition down the phylogenetic tree, posttranslational modifications, and substrate specificity suggests that PCPE2 acts as a dynamic regulator modulating both collagenous and noncollagenous pathways. Despite these recent advances, the mechanisms by which PCPE2 integrates its structural features into its function remain incompletely understood. Therefore, further investigation is needed to elucidate the role of PCPE2 in metabolic tissues and pathological conditions.

PCPE1 AND PCPE2: PHYSIOLOGICAL FUNCTION(S)

To look beyond the structural similarities and better understand the functional differences between PCPE1 and PCPE2, it is crucial to discuss recent studies using genetically engineered mouse models. Historically, PCPE1 expression has been shown to mirror that of collagen type I, which is upregulated in cultured fibroblasts⁴⁵ and in human and mouse organs undergoing progressive fibrosis, including liver, heart, skin, and cornea.^{45–50}

In addition, PCPE1 has been detected in human plasma, cerebrospinal fluid,^{51,52} and rodent plasma,^{18,46} suggesting that it may serve as a marker for muscle and liver fibrosis,¹⁸ while other studies suggest that plasma PCPE1 levels may serve as a biomarker for keratoconus,⁵³ a progressive disease that alters corneal remodeling. Furthering this idea, Massoudi et al⁵⁴ reported that PCPE1 plays a critical role in corneal homeostasis and scarring in mice. They also found that PCPE1 has antiangiogenic effects, suppressing pathological corneal neovascularization while promoting epithelial cell regeneration. In other studies, Luo et al⁵⁵ reported that the retinal vascular density from *Pcolce1* knockout (KO) mice was significantly lower than that of control mice, suggesting a role in retinal angiogenesis. Furthermore, silencing of *PCOLCE1* in human retinal microvascular endothelial cells inhibited cell proliferation and tube formation,⁵⁵ suggesting that PCPE1 facilitates angiogenesis by regulating collagen and modulating the ECM, thereby influencing vascular development.⁵⁵

In other investigations using global *Pcolce*^{KO} mice, abnormal collagen fibrils were noted in both mineralized and nonmineralized tissues, with 100% of tendon collagen fibrils exhibiting deranged morphologies,⁵⁶ altering bone geometry, and adaptive changes in bone size and shape.⁵⁶ Similarly, PCPE1 was reported to be significantly upregulated in human osteosarcoma and to promote tumor metastasis.⁵⁶ Wang et al⁵⁷ show that the PCPE1 expression was correlated with poor patient prognosis following examination of wild-type *PCOLCE* overexpression but not its N29Q mutant. These studies demonstrated that overexpression of PCPE1 promoted migration, invasion, and metastasis and suggested that PCPE1 glycosylation at Asn29 is necessary for its function in osteosarcoma. TWIST1, a key transcription factor in metastasis, was also overexpressed in osteosarcoma tissues and positively correlated with either PCPE1 or its potential procollagen substrates. These authors concluded that PCPE1 plays a critical role in promoting lung metastasis in osteosarcoma and may provide a therapeutic strategy for patients with metastatic osteosarcoma.⁵⁷ In another study, using global *Pcolce1*^{KO} mice exhibited significantly reduced liver fibrosis, body weight, and collagen I levels, yet did not show a reduction in the progression of nonalcoholic steatohepatitis.⁵⁸ These findings suggest that PCPE1 deficiency attenuates fibrosis while having minimal to no effect on liver steatosis. This same group in another published study showed that global *Pcolce1*^{KO} mice had significantly lower ejection fraction without any noted changes in cardiac collagen content or wound healing,⁵⁹ concluding that further studies should be conducted under stressed rather than basal conditions. In other mouse models utilizing global and tissue-specific PCPE1 mice, Hsiao et al^{60,61} reported that PCPE1 acts as a brown adipose tissue-derived cytokine that promotes liver fibrosis and

induces nonalcoholic steatohepatitis. In this study, both global and brown adipose tissue-specific deletions of PCPE1 markedly ameliorated liver fibrosis and circulating plasma PCPE1 levels. Conversely, overexpression of PCPE1 in brown adipose tissue exacerbated liver fibrosis, suggesting that high-calorie diet-induced stress increased PCPE1 production in brown adipose tissue, thereby increasing circulating PCPE1 levels and promoting liver fibrosis.⁶⁰ Taken together, these findings indicate that PCPE1 functions beyond its established role in BMP1 enhancement of collagen processing, acting as a secreted signaling molecule with cytokine-like properties that facilitate interorgan communication. Moreover, it also suggests that PCPE1 could be a multifunctional protein that links ECM remodeling with systemic signaling pathways to regulate fibrotic responses in other organs.

The number of physiological studies involving PCPE2 has been far fewer than for PCPE1. The first *Pcolce2* global KO mouse was reported to have normal thymic epithelial cell differentiation.⁶² Many years later, in 2009, PCPE2 was reported to enhance the cleavage of pro-apoA-I, a crucial step in the production of mature plasma apoA-1, the main protein constituent of HDL.^{4,63} Then, in 2011, the same group utilizing global *Pcolce2*^{KO} mice showed an approximately 2-fold higher plasma concentration of enlarged HDL particles in plasma, suggesting that the lack of PCPE2's enhancement of BMP1-mediated pro-apoA-I processing reduced cholesterol efflux capacity.^{5,64} To further investigate PCPE2's role in lipid metabolism, Pollard et al⁶⁵ crossed *Pcolce2*^{KO} mice with LDLR (low-density lipoprotein receptor) KO mice and observed enhanced progression of diet-induced atherosclerosis, which was related to reduced levels of liver SR-BI (scavenger receptor class B type 1)-mediated uptake of HDL.

PCPE2 has also been implicated in cardiac function.⁶⁶ In these studies, global *Pcolce2*^{KO} mice exhibited myocardial stiffness following transverse aortic constriction, suggesting that PCPE2 contributes to both myocardial compliance and collagen deposition in the mouse heart.⁶⁶ Beyond lipid metabolism and cardiac function, PCPE2 has been shown to enhance frontline immune defense.⁶⁷ In coculture experiments with tonsil-derived mesenchymal stem cells and differentiated HL-60 cells, knockdown of PCPE2 in the stem cells prevented the increase in reactive oxygen species production typically observed.⁶⁷ In other studies, adding recombinant PCPE2 increased reactive oxygen species production, highlighting its role in boosting neutrophil-like antimicrobial activity and suggesting that PCPE2 can be a crucial factor in enhancing frontline defense by generating reactive oxygen species to combat pathogens.⁶⁷ In another study, PCPE2 was linked to cellular plasticity and regeneration in the neonatal mouse cochlea.⁶⁸ In the neonatal mouse cochlea, overexpression of PCPE2 promoted reprogramming of supporting cells, enabling reentry into the

cell cycle and acquisition of hair cell-like properties,⁶⁸ suggesting that PCPE2 contributes to repair mechanisms and supports regenerative processes. In a study using the mouse digit tip as a robust model to study innate regeneration of composite tissues in mammals,⁶⁹ the authors found that overexpression of candidate profibrotic gene *Pcolce2* in the blastema modifies normal regeneration, and overexpression of candidate proregenerative factors in the proximal digit substantially increases bone deposition. These data demonstrate that the computational analysis combined with the AAV (adeno-associated virus) delivery approach provides a powerful framework for identifying the driving factors of fibrosis and regeneration in the mammalian digit.⁶⁹ In other studies involving type 2 diabetes and pancreatic islet cells, the *PCOLCE2* gene was identified as highly significant and was functionally tested.⁷⁰ Because type 2 diabetes is caused by insufficient insulin secretion from pancreatic β cells, genes were identified and tested, which appeared to contribute to this phenotype. The authors concluded that in type 2 diabetes, reduced insulin secretion is associated with a reduction in mitochondrial activity, and *PCOLCE2* expression is upregulated, contributing to impaired insulin secretion.⁷⁰ In studies involving adipose biology, PCPE2 expression was found to regulate lipid storage.³² In these studies, global *Pcolce2*^{KO} mice showed altered expression and function of SR-BI-mediated HDL uptake in adipocytes.³² While expression of PCPE2 enhances SR-BI-mediated HDL uptake, maintaining lipid balance in both adipose tissue and plasma, it also influences white adipose tissue expansion and inflammation. Loss of PCPE2 increased SR-BI expression but reduced adipocyte expansion and inflammation, indicating its critical role in adipose tissue homeostasis.³² Taken together, these findings illustrate that PCPE2 is a multifunctional protein with roles extending beyond ECM regulation. It is involved in lipid metabolism, cardiovascular function, innate immune defense, cellular plasticity, tissue regeneration, and adipose tissue homeostasis. The diverse functions of PCPE2 highlight its potential as a therapeutic target for metabolic, cardiovascular, and regenerative disorders, which underscores the need for further studies to fully investigate its mechanisms across different tissues and disease contexts.

CONCLUSIONS

Initially, PCPE1 and PCPE2 were identified as ECM proteins that regulate fibrillar collagen deposition. However, as their cellular functions have been explored in greater depth, it has become clear that both proteins have both related and distinct, multifaceted roles that extend well beyond collagen processing and ECM regulation. PCPE1 has emerged as an important mediator of fibrillogenesis and vascular processes. Rather than

acting solely as an autocrine factor that modulates cellular behavior through the ECM, PCPE1 also functions systemically as a secreted signaling molecule. Its involvement in brown adipose tissue-liver crosstalk, bone integrity, angiogenesis, liver fibrogenesis, and cancer metastasis highlights its capacity to link structural matrix regulation with paracrine and endocrine signaling pathways that govern tissue homeostasis and disease progression. In contrast, PCPE2 has been shown to have a more specific tissue distribution and a broader functional spectrum, encompassing cardiovascular physiology, metabolic regulation, cellular plasticity, and innate immune defense. PCPE2 plays a critical role in lipid metabolism by regulating apoA-I processing, HDL function, and SR-BI-mediated lipid uptake in both adipose tissue and the liver. Moreover, its involvement in myocardial stiffness and cochlear supporting cell reprogramming underscores its importance in cardiovascular health, regenerative capacity, and host defense mechanisms. Collectively, these emerging findings establish these interesting glycoproteins as multifunctional regulators that link ECM dynamics to metabolic, regenerative, and immune pathways. Further elucidation of their tissue-specific and context-dependent functions will be essential for assessing their therapeutic potential in metabolic, fibrotic, inflammatory, and cardiovascular diseases.

ARTICLE INFORMATION

Presented as the George Lyman Duff Memorial Lecture, New Orleans, LA, November 2025, as part of the American Heart Association Scientific Sessions. <https://www.ahajournals.org/atvb/lecture-series>.

Affiliations

Division of Endocrinology and Molecular Medicine, Department of Medicine (M.G.S.-T., B.A.), Cardiovascular Research Center (M.G.S.-T., A.R., B.A.), and Department of Pharmacology and Toxicology (M.G.S.-T., A.R.), Medical College of Wisconsin, Milwaukee.

Acknowledgments

I (Mary G. Sorci-Thomas) am incredibly honored to receive the 2025 George Lyman Duff Memorial Lecture. This award is a testament to the power of mentoring and collaborative science. I would like to dedicate this review to all my past and present mentees and collaborators: thank you for your trust, your curiosity, and the long hours spent in the laboratory, which helped push this research forward. You are the future of this field. I would also like to acknowledge my deepest respect for my late husband and collaborator, Michael Joseph Thomas, PhD. This work would not have been possible without his intellectual contributions, endless patience, and unwavering belief in this project. While he is no longer here to see it completed, his spirit and insights are on every page.

Source of Funding

The MST Laboratory is supported by the National Institutes of Health grant R01 DK139164.

Disclosures

None.

REFERENCES

1. Sorci-Thomas MG, Thomas MJ. Microdomains, inflammation, and atherosclerosis. *Circ Res*. 2016;118:679–691. doi: 10.1161/CIRCRESAHA.115.306246

2. Cheng HY, Gaddis DE, Wu R, McSkimming C, Haynes LD, Taylor AM, McNamara CA, Sorci-Thomas M, Hedrick CC. Loss of ABCG1 influences regulatory T cell differentiation and atherosclerosis. *J Clin Invest*. 2016;126:3236–3246. doi: 10.1172/JCI83136
3. Wilhelm AJ, Zabalawi M, Owen JS, Shah D, Grayson JM, Major AS, Bhat S, Gibbs DP Jr, Thomas MJ, Sorci-Thomas MG. Apolipoprotein A-I modulates regulatory T cells in autoimmune LDLR^{-/-}, ApoA-I^{-/-} mice. *J Biol Chem*. 2010;285:36158–36169. doi: 10.1074/jbc.M110.134130
4. Zhu J, Gardner J, Pullinger CR, Kane JP, Thompson JF, Francone OL. Regulation of apoA1 processing by procollagen C-proteinase enhancer-2 and bone morphogenetic protein-1. *J Lipid Res*. 2009;50:1330–1339. doi: 10.1194/jlr.M900034-JLR200
5. Francone OL, Ishida BY, de la Llera-Moya M, Royer L, Happe C, Zhu J, Chalkey RJ, Schaefer P, Cox C, Burlingame A, et al. Disruption of the murine procollagen C-proteinase enhancer 2 gene causes accumulation of pro-apoA-I and increased HDL levels. *J Lipid Res*. 2011;52:1974–1983. doi: 10.1194/jlr.M016527
6. Chang J, Pickard A, Herrera JA, O'Keefe S, Garva R, Hartshorn M, Hoyle A, Dingle L, Knox J, Jowitt TA, et al. Endocytic recycling is central to circadian collagen fibrillogenesis and disrupted in fibrosis. *Elife*. 2025;13:RP95842. doi: 10.7554/eLife.95842
7. Guillard J, Schworer S. Metabolic control of collagen synthesis. *Matrix Biol*. 2024;133:43–56. doi: 10.1016/j.matbio.2024.07.003
8. Lloyd SM, He Y. Exploring extracellular matrix crosslinking as a therapeutic approach to fibrosis. *Cells*. 2024;13:438. doi: 10.3390/cells13050438
9. Kisseleva T, Brenner DA. Mechanisms of fibrogenesis. *Exp Biol Med (Maywood)*. 2008;233:109–122. doi: 10.3181/0707-MR-190
10. Wynn TA, Ramalingam TR. Mechanisms of fibrosis: therapeutic translation for fibrotic disease. *Nat Med*. 2012;18:1028–1040. doi: 10.1038/nm.2807
11. Bourhis JM, Vadon-Le Goff S, Afrache H, Mariano N, Kronenberg D, Thielens N, Moali C, Hulmes DJ. Procollagen C-proteinase enhancer grasps the stalk of the C-propeptide trimer to boost collagen precursor maturation. *Proc Natl Acad Sci USA*. 2013;110:6394–6399. doi: 10.1073/pnas.1300480110
12. Moali C, Hulmes DJ. Extracellular and cell surface proteases in wound healing: new players are still emerging. *Eur J Dermatol*. 2009;19:552–564. doi: 10.1684/ejd.2009.0770
13. Muir A, Greenspan DS. Metalloproteinases in Drosophila to humans that are central players in developmental processes. *J Biol Chem*. 2011;286:41905–41911. doi: 10.1074/jbc.R111.299768
14. Napoli M, Bauer J, Bonod C, Goff SVL, Moali C. PCPE-2 (procollagen C-proteinase enhancer-2): The non-identical twin of PCPE-1. *Matrix Biol*. 2024;134:59–78. doi: 10.1016/j.matbio.2024.09.001
15. Thomas MJ, Xu H, Wang AEL, Beg MA, Sorci-Thomas MG. PCPE2: Expression of multifunctional extracellular glycoprotein associated with diverse cellular functions. *J Lipid Res*. 2024;65:100664. doi: 10.1016/j.jlcr.2024.100664
16. Vadon-Le Goff S, Tessier A, Napoli M, Dieryckx C, Bauer J, Dussoyer M, Lagoutte P, Peyronnel C, Essayan L, Kleiser S, et al. Identification of PCPE-2 as the endogenous specific inhibitor of human BMP-1/tolloid-like proteinases. *Nat Commun*. 2023;14:8020. doi: 10.1038/s41467-023-43401-0
17. Lagoutte P, Bettler E, Vadon-Le Goff S, Moali C. Procollagen C-proteinase enhancer-1 (PCPE-1), a potential biomarker and therapeutic target for fibrosis. *Matrix Biol Plus*. 2021;1:11:100062. doi: 10.1016/j.mbplus.2021.100062
18. Hassoun E, Safrin M, Ziv H, Pri-Chen S, Kessler E. Correction: procollagen C-proteinase enhancer 1 (PCPE-1) as a plasma marker of muscle and liver fibrosis in mice. *PLoS One*. 2016;11:e0162747. doi: 10.1371/journal.pone.0162747
19. Kessler E, Mould AP, Hulmes DJ. Procollagen type I C-proteinase enhancer is a naturally occurring connective tissue glycoprotein. *Biochem Biophys Res Commun*. 1990;173:81–86. doi: 10.1016/s0006-291x(05)81024-1
20. Vadon-Le Goff S, Kronenberg D, Bourhis JM, Bijakowski C, Raynal N, Ruggiero F, Farndale RW, Stocker W, Hulmes DJ, Moali C. Procollagen C-proteinase enhancer stimulates procollagen processing by binding to the C-propeptide region only. *J Biol Chem*. 2011;286:38932–38938. doi: 10.1074/jbc.M111.274944
21. Pulido D, Sharma U, Vadon-Le Goff S, Hussain SA, Cordes S, Mariano N, Bettler E, Moali C, Aghajari N, Hohenester E, et al. Structural basis for the acceleration of procollagen processing by procollagen C-proteinase enhancer-1. *Structure*. 2018;26:1384–1392.e3. doi: 10.1016/j.str.2018.06.011
22. Bork P, Beckmann G. The CUB domain. A widespread module in developmentally regulated proteins. *J Mol Biol*. 1993;231:539–545. doi: 10.1006/jmbi.1993.1305
23. Major B, Kardos J, Kekesi KA, Lorincz Z, Zavodszky P, Gal P. Calcium-dependent conformational flexibility of a CUB domain controls activation of the complement serine protease C1r. *J Biol Chem*. 2010;285:11863–11869. doi: 10.1074/jbc.M109.098541
24. Xu H, Acott TS, Wirtz MK. Identification and expression of a novel type I procollagen C-proteinase enhancer protein gene from the glaucoma candidate region on 3q21-q24. *Genomics*. 2000;66:264–273. doi: 10.1006/geno.2000.6229
25. Weiss T, Ricard-Blum S, Moschovich L, Wineman E, Mesilaty S, Kessler E. Binding of procollagen C-proteinase enhancer-1 (PCPE-1) to heparin/heparan sulfate: properties and role in PCPE-1 interaction with cells. *J Biol Chem*. 2010;285:33867–33874. doi: 10.1074/jbc.M110.141366
26. Weiss T, Brusel M, Rousselle P, Kessler E. The NTR domain of procollagen C-proteinase enhancer-1 (PCPE-1) mediates PCPE-1 binding to syndecans-1, -2 and -4 as well as fibronectin. *Int J Biochem Cell Biol*. 2014;57:45–53. doi: 10.1016/j.biocel.2014.09.023
27. Bekhouche M, Kronenberg D, Vadon-Le Goff S, Bijakowski C, Lim NH, Font B, Kessler E, Colige A, Nagase H, Murphy G, et al. Role of the netrin-like domain of procollagen C-proteinase enhancer-1 in the control of metalloproteinase activity. *J Biol Chem*. 2010;285:15950–15959. doi: 10.1074/jbc.M109.086447
28. Consortium GT. The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science*. 2020;369:1318–1330. doi: 10.1126/science.aaz1776
29. Steiglitz BM, Keene DR, Greenspan DS. PCOLCE2 encodes a functional procollagen C-proteinase enhancer (PCPE2) that is a collagen-binding protein differing in distribution of expression and post-translational modification from the previously described PCPE1. *J Biol Chem*. 2002;277:49820–49830. doi: 10.1074/jbc.M209891200
30. Mesilaty-Gross S, Anikster Y, Vilensky B, Wolf I, Phillip M, Gat-Yablonski G. Different patterns of human serum procollagen C-proteinase enhancer1 (PCPE1). *Clin Chim Acta*. 2009;403:76–80. doi: 10.1016/j.cca.2009.01.024
31. Ma L, Gan C, Huang Y, Wang Y, Luo G, Wu J. Comparative proteomic analysis of extracellular matrix proteins secreted by hypertrophic scar with normal skin fibroblasts. *Burns Trauma*. 2014;2:76–83. doi: 10.4103/2321-3868.130191
32. Xu H, Thomas MJ, Kaul S, Kallinger R, Ouweneel AB, Maruko E, Oussada SM, Jongejan A, Cense HA, Nieuwdorp M, et al. Pcp2, a novel extracellular matrix protein, regulates adipocyte SR-BI-mediated high-density lipoprotein uptake. *Arterioscler Thromb Vasc Biol*. 2021;41:2708–2725. doi: 10.1161/ATVBAHA.121.316615
33. Pollard RD, Blesso CN, Zabalawi M, Fulp B, Gerelus M, Zhu X, Lyons EW, Nuradin N, Francone OL, Li XA, et al. Procollagen C-endopeptidase enhancer protein 2 (PCPE2) reduces atherosclerosis in mice by enhancing scavenger receptor class B1 (SR-BI)-mediated high-density lipoprotein (HDL)-cholesteryl ester uptake. *J Biol Chem*. 2015;290:15496–15511. doi: 10.1074/jbc.M115.646240
34. Salza R, Peysselon F, Chautard E, Faye C, Moschovich L, Weiss T, Perrin-Cocon L, Lotteau V, Kessler E, Ricard-Blum S. Extended interaction network of procollagen C-proteinase enhancer-1 in the extracellular matrix. *Biochem J*. 2014;457:137–149. doi: 10.1042/BJ20130295
35. Adar R, Kessler E, Goldberg B. Evidence for a protein that enhances the activity of type I procollagen C-proteinase. *Coll Relat Res*. 1986;6:267–277. doi: 10.1016/s0174-173x(86)80011-5
36. Bernocco S, Steiglitz BM, Svergun DI, Petoukhov MV, Ruggiero F, Ricard-Blum S, Ebel C, Geourjon C, Deleage G, Font B, et al. Low resolution structure determination shows procollagen C-proteinase enhancer to be an elongated multidomain glycoprotein. *J Biol Chem*. 2003;278:7199–7205. doi: 10.1074/jbc.M210857200
37. Takahara K, Kessler E, Biniaminov L, Brusel M, Eddy RL, Jani-Sait S, Shows TB, Greenspan DS. Type I procollagen COOH-terminal proteinase enhancer protein: identification, primary structure, and chromosomal localization of the cognate human gene (PCOLCE). *J Biol Chem*. 1994;269:26280–26285. doi: https://doi.org/10.1016/S0021-9258(18)47191-8
38. Kronenberg D, Vadon-Le Goff S, Bourhis JM, Font B, Eichenberger D, Hulmes DJ, Moali C. Strong cooperativity and loose geometry between CUB domains are the basis for procollagen c-proteinase enhancer activity. *J Biol Chem*. 2009;284:33437–33446. doi: 10.1074/jbc.M109.046128
39. Moali C, Font B, Ruggiero F, Eichenberger D, Rousselle P, Francois V, Oldberg A, Bruckner-Tuderman L, Hulmes DJ. Substrate-specific modulation of a multisubstrate proteinase. C-terminal processing of fibrillar procollagens is the only BMP-1-dependent activity to be enhanced by PCPE-1. *J Biol Chem*. 2005;280:24188–24194. doi: 10.1074/jbc.M501486200
40. Kessler E, Adar R. Type I procollagen C-proteinase from mouse fibroblasts. Purification and demonstration of a 55-kDa enhancer glycoprotein. *Eur J Biochem*. 1989;186:115–121. doi: 10.1111/j.1432-1033.1989.tb15184.x

41. Hornbeck PV, Zhang B, Murray B, Kornhauser JM, Latham V, Skrzypek E. PhosphoSitePlus, 2014: mutations, PTMs and recalibrations. *Nucleic Acids Res.* 2015;43:D512–D520. doi: 10.1093/nar/gku1267
42. Mayya V, Lundgren DH, Hwang SI, Rezaul K, Wu L, Eng JK, Rodionov V, Han DK. Quantitative phosphoproteomic analysis of T cell receptor signaling reveals system-wide modulation of protein-protein interactions. *Sci Signal.* 2009;2:ra46. doi: 10.1126/scisignal.2000007
43. Sui S, Wang J, Yang B, Song L, Zhang J, Chen M, Liu J, Lu Z, Cai Y, Chen S, et al. Phosphoproteome analysis of the human Chang liver cells using SCX and a complementary mass spectrometric strategy. *Proteomics.* 2008;8:2024–2034. doi: 10.1002/pmic.200700896
44. Mertins P, Yang F, Liu T, Mani DR, Petyuk VA, Gillette MA, Clauser KR, Qiao JW, Gritsenko MA, Moore RJ, et al. Ischemia in tumors induces early and sustained phosphorylation changes in stress kinase pathways but does not affect global protein levels. *Mol Cell Proteomics.* 2014;13:1690–1704. doi: 10.1074/mcp.M113.036392
45. Ogata I, Auster AS, Matsui A, Greenwel P, Geerts A, D'Amico T, Fujiwara K, Kessler E, Rojkind M. Up-regulation of type I procollagen C-proteinase enhancer protein messenger RNA in rats with CCl4-induced liver fibrosis. *Hepatology.* 1997;26:611–617. doi: 10.1002/hep.510260312
46. Ippolito DL, AbdulHameed MD, Tawa GJ, Baer CE, Permenter MG, McDyre BC, Dennis WE, Boyle MH, Hobbs CA, Streicker MA, et al. Gene expression patterns associated with histopathology in toxic liver fibrosis. *Toxicol Sci.* 2016;149:67–88. doi: 10.1093/toxsci/kfv214
47. Kessler-Icekson G, Schlesinger H, Freimann S, Kessler E. Expression of procollagen C-proteinase enhancer-1 in the remodeling rat heart is stimulated by aldosterone. *Int J Biochem Cell Biol.* 2006;38:358–365. doi: 10.1016/j.biocel.2005.10.007
48. Yu L, Ruifrok WP, Meissner M, Bos EM, van Goor H, Sanjabi B, van der Harst P, Pitt B, Goldstein IJ, Koerts JA, et al. Genetic and pharmacological inhibition of galectin-3 prevents cardiac remodeling by interfering with myocardial fibrogenesis. *Circ Heart Fail.* 2013;6:107–117. doi: 10.1161/CIRCHEARTFAILURE.112.971168
49. Wong VW, You F, Janusz M, Gurtner GC, Kuang AA. Transcriptional profiling of rapamycin-treated fibroblasts from hypertrophic and keloid scars. *Ann Plast Surg.* 2014;72:711–719. doi: 10.1097/SAP.0b013e31826956f6
50. Malecize F, Massoudi D, Fournie P, Tricoire C, Cassagne M, Malbouyres M, Hulmes DJ, Moali C, Galiacy SD. Upregulation of bone morphogenetic protein-1/mammalian tolloid and procollagen C-proteinase enhancer-1 in corneal scarring. *Invest Ophthalmol Vis Sci.* 2014;55:6712–6721. doi: 10.1167/iov.13-13800
51. Finehout EJ, Franck Z, Lee KH. Towards two-dimensional electrophoresis mapping of the cerebrospinal fluid proteome from a single individual. *Electrophoresis.* 2004;25:2564–2575. doi: 10.1002/elps.200406012
52. Yuan X, Desiderio DM. Protein identification with Teflon as matrix-assisted laser desorption/ionization sample support. *J Mass Spectrom.* 2002;37:512–524. doi: 10.1002/jms.307
53. Saglam ZG, Yildirim M, Laloglu E. Evaluating serum biomarkers in keratoconus: the role of PCPE-1 and PCPE-2 in collagen metabolism and disease progression. *Graefes Arch Clin Exp Ophthalmol.* 2025;263:2841–2848. doi: 10.1007/s00417-025-06910-w
54. Massoudi D, Germer CJ, Glisch JM, Greenspan DS. Procollagen C-proteinase enhancer 1 (PCPE-1) functions as an anti-angiogenic factor and enhances epithelial recovery in injured cornea. *Cell Tissue Res.* 2017;370:461–476. doi: 10.1007/s00441-017-2689-6
55. Luo J, Zhao PQ, Chen HJ, Liu MM, He JQ, Fei P. Procollagen C-proteinase enhancer 1 promotes physiologic retinal angiogenesis via regulating the process of collagen. *Int J Ophthalmol.* 2022;15:868–875. doi: 10.18240/ijo.2022.06.03
56. Steiglitiz BM, Kreider JM, Frankenburg EP, Pappano WN, Hoffman GG, Meganck JA, Liang X, Hook M, Birk DE, Goldstein SA, et al. Procollagen C proteinase enhancer 1 genes are important determinants of the mechanical properties and geometry of bone and the ultrastructure of connective tissues. *Mol Cell Biol.* 2006;26:238–249. doi: 10.1128/MCB.26.1.238-249.2006
57. Wang S, Zhong L, Li Y, Xiao D, Zhang R, Liao D, Lv D, Wang X, Wang J, Xie X, et al. Up-regulation of PCOLCE by TWIST1 promotes metastasis in osteosarcoma. *Theranostics.* 2019;9:4342–4353. doi: 10.7150/thno.34090
58. Sansilvestri Morel P, Duvivier V, Bertin F, Provost N, Hammoutene A, Hubert EL, Gonzalez A, Tupinon-Mathieu I, Paradis V, Delerive P. Procollagen C-proteinase enhancer-1 (PCPE-1) deficiency in mice reduces liver fibrosis but not NASH progression. *PLoS One.* 2022;17:e0263828. doi: 10.1371/journal.pone.0263828
59. Sansilvestri-Morel P, Harouki-Crochemore N, Bertin F, Bertheux H, Vermeil de Conchard G, Digue N, Desfosses E, Lecomte M, Gonzalez A, Diez J, et al. Deficiency of procollagen C-proteinase enhancer 1 in mice has no major impact on cardiac collagen and function under basal conditions. *J Cardiovasc Pharmacol.* 2021;78:e703–e713. doi: 10.1097/FJC.0000000000001122
60. Hsiao YT, Yoshida Y, Okuda S, Abe M, Mizuno S, Takahashi S, Nakagami H, Morishita R, Kamimura K, Terai S, et al. PCPE-1, a brown adipose tissue-derived cytokine, promotes obesity-induced liver fibrosis. *EMBO J.* 2024;43:4846–4869. doi: 10.1038/s44318-024-00196-0
61. Pereira RO, Abel ED. BATokines in metabolic liver disease: good cops or bad cops?. *EMBO J.* 2024;43:4807–4809. doi: 10.1038/s44318-024-00239-6
62. Heinzel K, Bleul CC. The Foxn1-dependent transcripts PCOLCE2 and mPPP1R16B are not required for normal thymopoiesis. *Eur J Immunol.* 2007;37:2562–2571. doi: 10.1002/eji.200637474
63. Sviridov D. Maturation of apolipoprotein A-I: unrecognized health benefit or a forgotten rudiment?. *J Lipid Res.* 2009;50:1257–1258. doi: 10.1194/jlr.E900003-JLR200
64. Getz GS, Reardon CA. Cubbing in proapolipoprotein maturation. *J Lipid Res.* 2011;52:1861–1864. doi: 10.1194/jlr.R020149
65. Pollard RD, Fulp B, Sorci-Thomas MG, Thomas MJ. High-density lipoprotein biogenesis: defining the domains involved in human apolipoprotein A-I lipidation. *Biochemistry.* 2016;55:4971–4981. doi: 10.1021/acs.biochem.6b00347
66. Baicu CF, Zhang YH, Van Laer AO, Renaud L, Zile MR, Bradshaw AD. Effects of the absence of procollagen C-endopeptidase enhancer-2 on myocardial collagen accumulation in chronic pressure overload. *Am J Physiol Heart Circ Physiol.* 2012;303:H234–H240. doi: 10.1152/ajpheart.00227.2012
67. Yoon HS, Kim HY, Cho KA, Kim YH, Woo SY, Kim HS, Kang JL, Ryu KH, Park JW. Procollagen C-endopeptidase enhancer 2 secreted by tonsil-derived mesenchymal stem cells increases the oxidative burst of promyelocytic HL-60 cells. *Biology (Basel).* 2022;11:255. doi: 10.3390/biology11020255
68. Xu C, Zhang L, Zhou Y, Du H, Qi J, Tan F, Peng L, Gu X, Li N, Sun Q, et al. Pcolce2 overexpression promotes supporting cell reprogramming in the neonatal mouse cochlea. *Cell Prolif.* 2024;57:e13633. doi: 10.1111/cpr.13633
69. Jou V, Semelsberger SD, Smerczynski J, Lehoczy JA. Mouse digit AAV gene delivery into fibroblasts regulates regenerative outcome. *Sci Adv.* 2026;12:eadz0229. doi: 10.1126/sciadv.adz0229
70. Bacos K, Perilyev A, Karagiannopoulos A, Cowan E, Ofori JK, Bertonnier-Brouty L, Ronn T, Lindqvist A, Luan C, Ruhmann S, et al. Type 2 diabetes candidate genes, including PAX5, cause impaired insulin secretion in human pancreatic islets. *J Clin Invest.* 2023;133:e163612. doi: 10.1172/JCI163612